AI OPEN
DATA CENTER



Enabling Grid-Compliant Data Center Growth Through Advanced Energy Storage

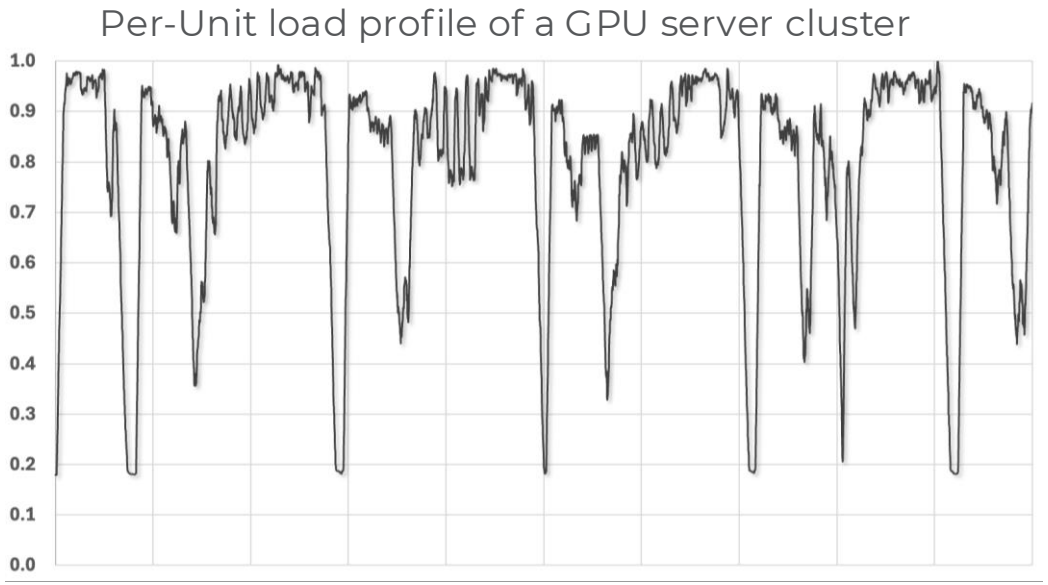
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AI Load is a Grid Risk

- Large load trips
  - 1.5GW (2024), 1.8GW (2025), 3GW (2026)
- Multi-megawatt step changes in seconds, not minutes
  - Different from the flat, predictable load
- Sub-scynchronous oscillations (SSO) can damage nearby generation
  - Need for active damping
- Utilities are responding by protecting grid stability with new requirements



Challenges → New Requirements

- **Voltage ride through** – Detailed dynamic load models, disconnect/reconnect thresholds, system voltage ride-through and load transfer capabilities, behind the meter resources
- **Load variability**– limits on power swings, flicker, and harmonic distortion
  - Ramp rate limits (MW/min or MW/sec)
- **Sub-synchronous oscillation** – dampen response between power electronic loads, BESS active power compensation, etc.
- New requirements from the following entities:
  - NERC, ERCOT, PJM, MISO, SPP, CAISO, ISO-NE, NYISO, AEP Ohio, Georgia Power, Dominion Energy Virginia, Duke Energy, Xcel Energy

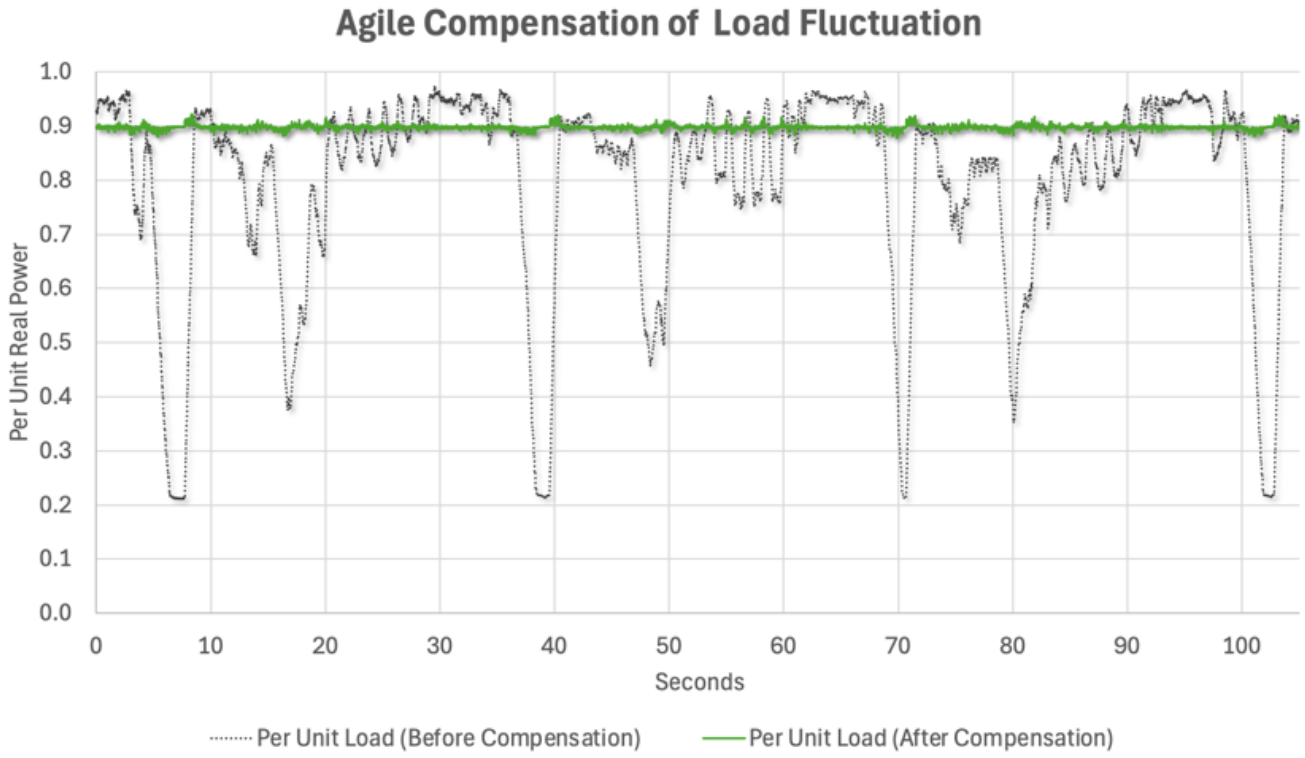
Emerging Interconnection Requirements

	妖 Explicit/Codified		坦 Study-based		Not specified	Proposed		
Requirement	ercot	SPP Southwest Power Pool	MISO	ATC	Southern Company	LIPA	aeso	PG&E
Explicit VRT Envelope	Yes (NOGRR 282)	Yes (HILL)	Yes proposed	Yes (Sec. 9)	Review and approve	Yes	Yes (TCDC)	Yes
Explicit FRT Envelope	Yes 299 s bands	Yes (PRC-029)	Yes proposed	Study (no FRT)	Review and approve	Yes 299 s	Yes 60 s	Yes 299 s bands
PFAPR (≥90% rule)	90% 1 s	90% 1 s	90% 1 s	Study based	Review and approve	90% 1 s	90% config.	90% 2–3 s ★
Ramp Rate Limit (Minute-Level)	Study based	20 MW /min	30 MW /min	30 MW /min. equivalent.	20 MW /min	20 MW /min	10 MW /min	20 MW /min
Ramp Rate Limit (Sub-Minute)	10 MW 5 s proposed	25 MW 5 s	10 MW 5 s proposed	25 MW 5 s	10 MW 4–6 s	50% P 0.1 s	Study	10 MW 4–6 s
Oscillation Mitigation	10 MW 5 s proposed	EMT screening	Study PSPD+EMT	SSO analysis	4 bands FFT+PMU	2 bands PMU	EMT study	Study 2 bands
Reactive Power / Power Factor	None specified	±0.95 PF at POI	Study based	±0.95 PF	±0.90 PF	±0.95 PF	±0.95 PF	0.97-0.99 (±0.99 if voltage problem found)
Power Quality (IEEE 519)	Study	Yes	Yes proposed	Yes	Yes	Yes	Yes	Yes (IEEE 519-2022)
Monitoring (PMU/DFR)	DFR required	DFR	DFR proposed	DFR	PMU required	SCADA + monitor	PMU at POC	PMU/DFR + telemetry

Source: Danovo Energy Solutions, a Quanta Services company

The Solution: Energy Storage as a Flexible Buffer

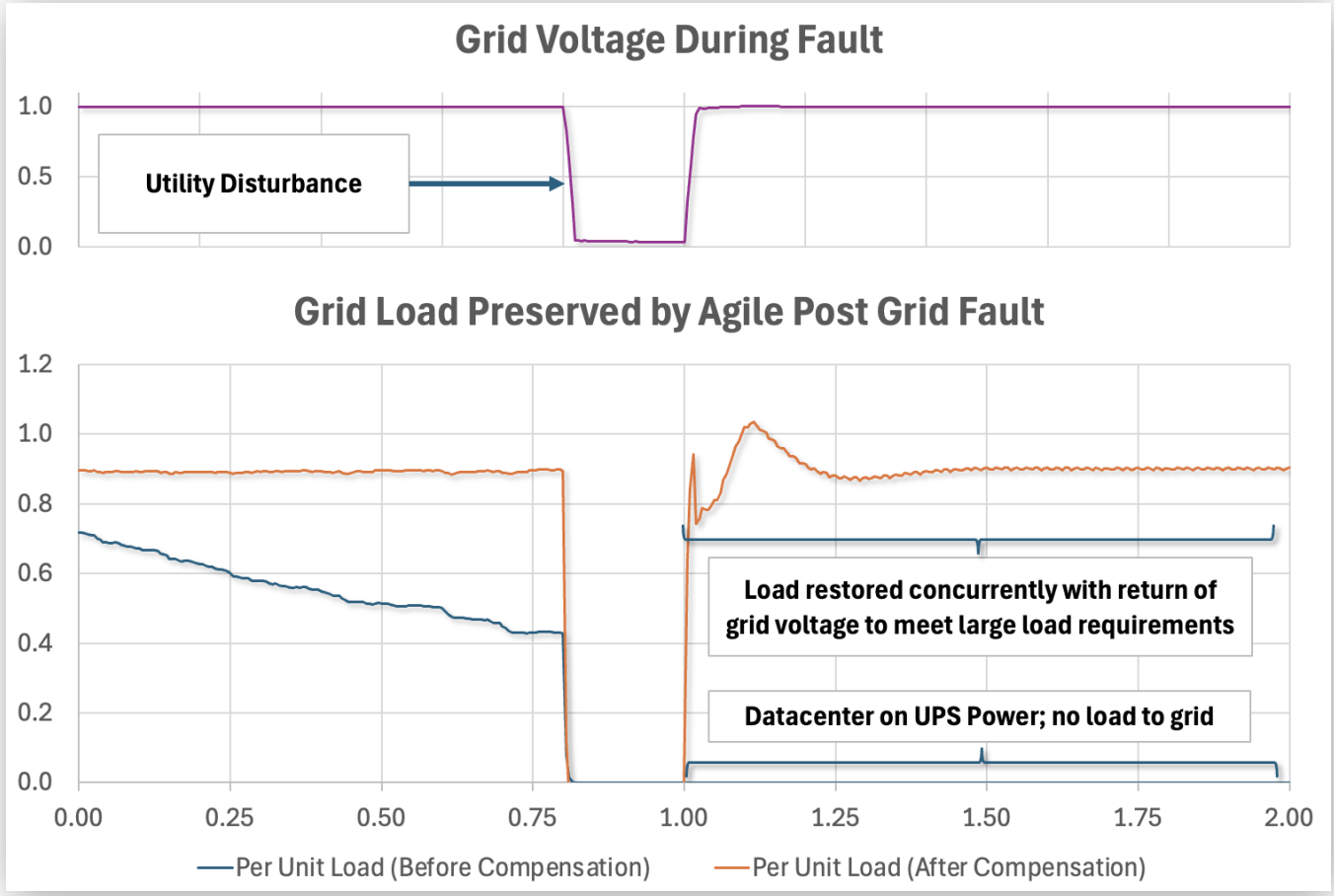
- Energy Storage Systems (ESS) can:
  - Buffer the AI load for the grid
  - Enable grid ride-through compliance
  - Provide black start capability
  - Improved power quality
- Presents a stable, compliant interface at the POI
- Easy to implement from a controls anc electrical interconnection
- Flexibility

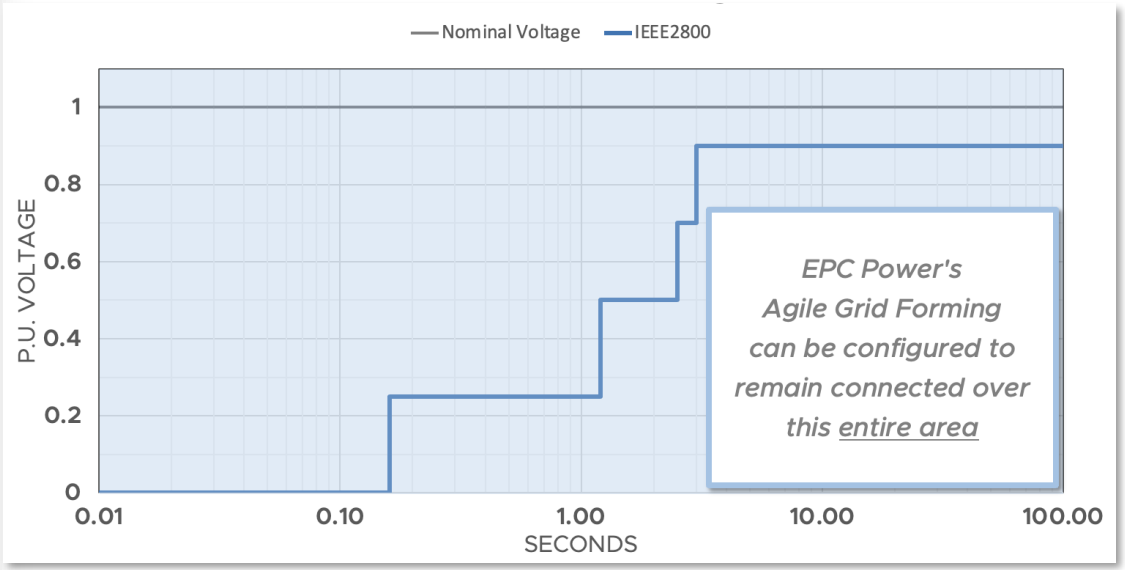


Need Advanced Power Conversion

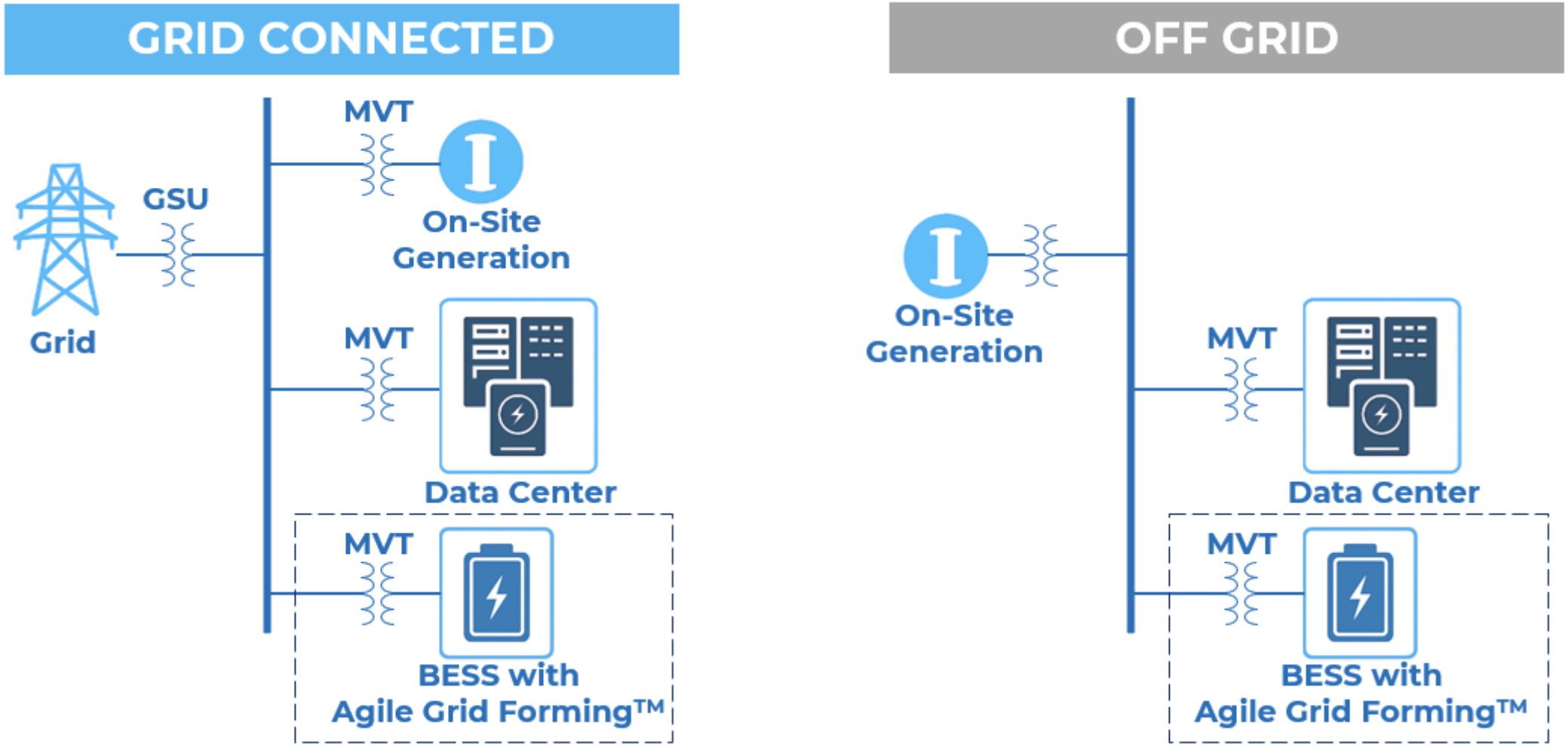
- Initial grid forming response is nearly instantaneous, but isn't sufficient
- The larger the site, the stronger the grid, the harder to comply without advanced controls
- Standard ESS controls respond in sub-second timeframes
- Advanced ESS controls (e.g. Agile Grid Forming™) respond in sub-cycle timeframes
- ESS detects load steps and transients and reacts before they propagate to the point of interconnection

How to Approach Ride-Through



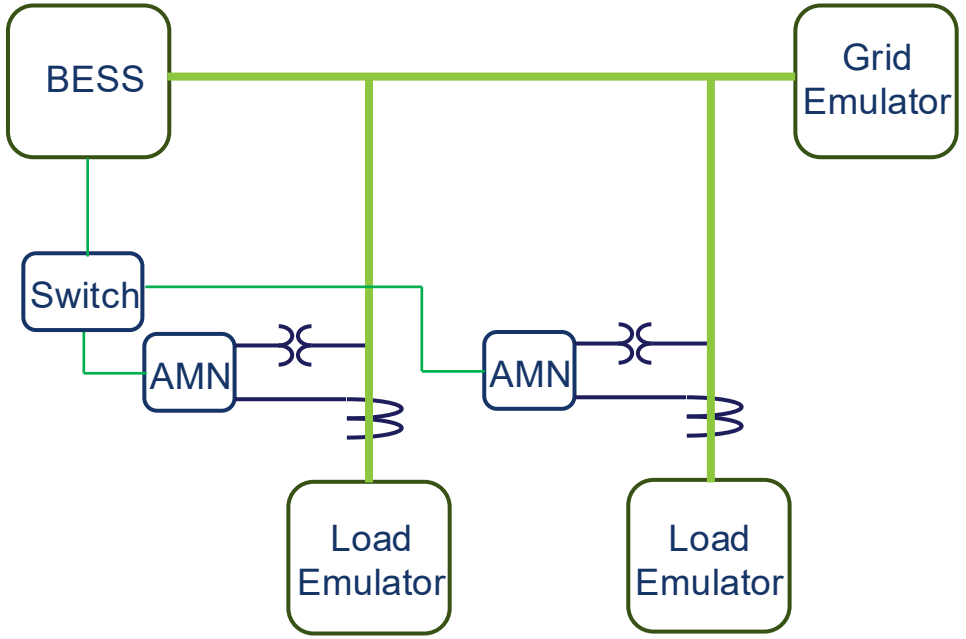


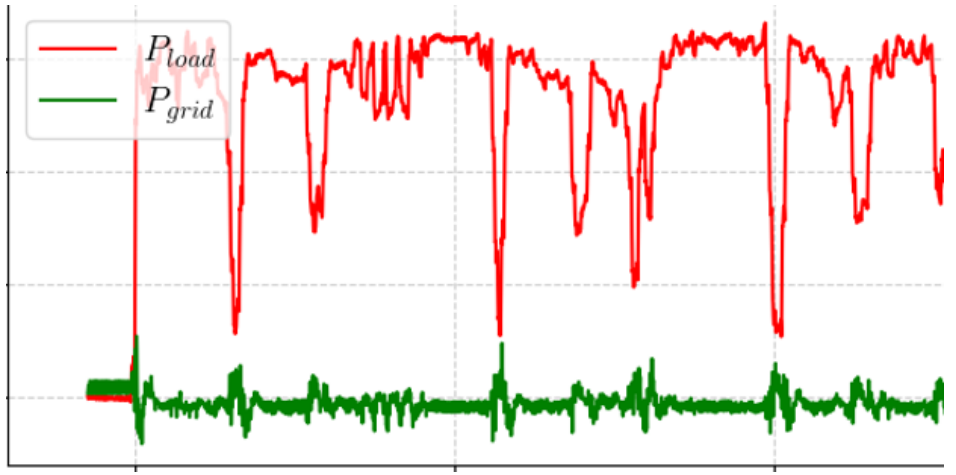
Integration with On-Site Generation



Model and Hardware Testing

- Validation testing is essential to demonstrate compliance with EMT models for utility interconnection requirements





Key Takeaways

- Ramp-rate limits, ride-through, and oscillation damping controls are now standard interconnection requirements for AI data centers — not edge cases
- NVIDIA's DSX architecture includes ESS in all stages
- Energy storage is the flexible buffer that lets IT load stay dynamic while the grid sees a stable, compliant, predictable interface
- Engaging storage design early in the interconnection process reduces risk, shortens approval timelines, and protects growth plans
- Call to action: get involved in industry organizations trying to have discussions to quantify and validate the solutions that exist. OCP, NERC LLWG, ERCOT LLWG, etc.

Call to Action

- Get involved in industry organizations trying to have discussions to quantify and validate the solutions that exist and the applications for them in data centers
  - OCP Energy Storage Systems Workstream
  - NERC LLWG, ERCOT LLWG, etc.

Thank You!

